

VESSEL ANALYSIS REPORT

vaf(%)	branch points count	tip point count	vein count	total vein length	average vein length
1.659	83.0	207.0	226.0	4769.955	21.106

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
1	[54,4]	[64,10]	11.485	BRANCH	BRANCH
2	[2,4]	[54,4]	54.728	TIP	BRANCH
3	[54,4]	[57,17]	14.828	BRANCH	BRANCH
4	[251,6]	[248,7]	2.414	TIP	BRANCH
5	[246,5]	[248,7]	2.414	TIP	BRANCH
6	[248,7]	[248,9]	1.0	BRANCH	BRANCH
7	[83,0]	[81,9]	8.828	TIP	BRANCH
8	[106,34]	[81,9]	37.284	TIP	BRANCH
9	[81,9]	[64,10]	16.414	BRANCH	BRANCH
10	[248,9]	[237,10]	11.243	BRANCH	BRANCH
11	[252,30]	[248,9]	21.243	TIP	BRANCH
12	[64,10]	[57,17]	10.657	BRANCH	BRANCH
13	[216,0]	[204,10]	17.485	TIP	BRANCH
14	[169,0]	[204,10]	38.142	TIP	BRANCH
15	[197,18]	[204,10]	11.657	TIP	BRANCH
16	[245,0]	[237,10]	12.899	TIP	BRANCH
17	[228,23]	[237,10]	16.899	TIP	BRANCH
18	[49,26]	[57,17]	13.071	TIP	BRANCH
19	[453,24]	[456,25]	2.414	TIP	BRANCH
20	[458,28]	[456,25]	3.414	TIP	BRANCH
21	[493,1]	[456,25]	49.042	TIP	BRANCH
22	[140,26]	[141,33]	6.414	TIP	BRANCH
23	[141,33]	[161,39]	27.142	BRANCH	BRANCH
24	[141,33]	[121,46]	25.142	BRANCH	BRANCH
25	[1,25]	[8,36]	14.071	TIP	BRANCH
26	[28,26]	[8,36]	24.314	TIP	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
27	[0,45]	[8,36]	12.071	TIP	BRANCH
28	[161,39]	[180,53]	24.971	BRANCH	BRANCH
29	[161,39]	[124,64]	52.284	BRANCH	BRANCH
30	[119,46]	[121,46]	1.0	BRANCH	BRANCH
31	[119,46]	[96,49]	23.657	BRANCH	BRANCH
32	[119,46]	[99,74]	34.87	BRANCH	BRANCH
33	[121,46]	[124,64]	17.828	BRANCH	BRANCH
34	[91,41]	[96,49]	9.071	TIP	BRANCH
35	[84,61]	[96,49]	16.142	TIP	BRANCH
36	[180,53]	[188,54]	7.414	BRANCH	BRANCH
37	[179,56]	[180,53]	2.414	TIP	BRANCH
38	[227,32]	[193,53]	42.284	TIP	BRANCH
39	[193,53]	[188,54]	4.414	BRANCH	BRANCH
40	[215,63]	[193,53]	25.728	TIP	BRANCH
41	[188,54]	[209,84]	38.284	BRANCH	BRANCH
42	[371,50]	[370,56]	5.414	TIP	BRANCH
43	[334,95]	[370,56]	58.184	TIP	BRANCH
44	[371,58]	[370,56]	1.414	TIP	BRANCH
45	[395,0]	[319,62]	103.61	TIP	BRANCH
46	[319,62]	[306,71]	16.899	BRANCH	BRANCH
47	[318,60]	[319,62]	1.0	TIP	BRANCH
48	[124,64]	[114,81]	22.071	BRANCH	BRANCH
49	[314,50]	[306,71]	43.556	TIP	BRANCH
50	[306,71]	[293,99]	32.385	BRANCH	BRANCH
51	[106,71]	[99,74]	7.243	TIP	BRANCH
52	[99,74]	[85,78]	14.657	BRANCH	BRANCH
53	[86,71]	[85,78]	6.414	TIP	BRANCH
54	[85,78]	[81,88]	10.657	BRANCH	BRANCH
55	[85,78]	[81,88]	13.657	BRANCH	BRANCH
56	[114,81]	[115,106]	28.556	BRANCH	BRANCH
57	[110,78]	[114,81]	3.828	TIP	BRANCH
58	[222,92]	[209,84]	15.314	TIP	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
59	[201,85]	[209,84]	7.414	TIP	BRANCH
60	[39,81]	[37,85]	3.828	TIP	BRANCH
61	[37,85]	[15,97]	25.971	BRANCH	BRANCH
62	[44,99]	[37,85]	16.314	TIP	BRANCH
63	[81,88]	[71,108]	23.142	BRANCH	BRANCH
64	[2,61]	[12,89]	39.042	TIP	BRANCH
65	[12,89]	[15,97]	8.243	BRANCH	BRANCH
66	[10,85]	[12,89]	3.414	TIP	BRANCH
67	[15,97]	[21,116]	20.485	BRANCH	BRANCH
68	[241,60]	[293,99]	70.669	TIP	BRANCH
69	[293,99]	[307,112]	19.728	BRANCH	BRANCH
70	[359,93]	[353,104]	12.485	TIP	BRANCH
71	[353,104]	[309,112]	47.971	BRANCH	BRANCH
72	[447,159]	[353,104]	120.054	TIP	BRANCH
73	[115,106]	[115,115]	8.0	BRANCH	BRANCH
74	[140,84]	[115,106]	36.799	TIP	BRANCH
75	[71,108]	[64,111]	6.828	BRANCH	BRANCH
76	[71,108]	[77,116]	9.071	BRANCH	BRANCH
77	[64,111]	[48,122]	21.314	BRANCH	BRANCH
78	[61,110]	[64,111]	2.0	TIP	BRANCH
79	[307,112]	[309,112]	1.0	BRANCH	BRANCH
80	[307,112]	[362,159]	83.497	BRANCH	BRANCH
81	[318,121]	[309,112]	11.899	TIP	BRANCH
82	[115,115]	[93,119]	22.657	BRANCH	BRANCH
83	[115,115]	[122,119]	7.243	BRANCH	BRANCH
84	[21,116]	[23,116]	1.0	BRANCH	BRANCH
85	[16,122]	[21,116]	6.657	TIP	BRANCH
86	[33,107]	[23,116]	12.314	TIP	BRANCH
87	[35,130]	[23,116]	17.556	TIP	BRANCH
88	[77,116]	[93,119]	16.243	BRANCH	BRANCH
89	[77,116]	[75,135]	18.828	BRANCH	BRANCH
90	[157,91]	[122,119]	46.77	TIP	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
91	[119,132]	[122,119]	13.243	TIP	BRANCH
92	[48,122]	[75,135]	34.899	BRANCH	BRANCH
93	[182,122]	[209,131]	37.385	TIP	BRANCH
94	[209,131]	[216,131]	6.0	BRANCH	BRANCH
95	[199,138]	[209,131]	12.485	TIP	BRANCH
96	[225,138]	[216,131]	11.485	TIP	BRANCH
97	[280,108]	[216,131]	73.284	TIP	BRANCH
98	[75,137]	[75,135]	1.0	TIP	BRANCH
99	[18,155]	[35,162]	30.556	BRANCH	BRANCH
100	[12,126]	[18,155]	41.456	TIP	BRANCH
101	[18,155]	[19,159]	3.414	BRANCH	BRANCH
102	[19,159]	[35,162]	16.243	BRANCH	BRANCH
103	[19,159]	[18,167]	7.0	BRANCH	BRANCH
104	[362,154]	[362,159]	4.0	TIP	BRANCH
105	[373,162]	[362,159]	11.243	TIP	BRANCH
106	[35,162]	[57,174]	29.657	BRANCH	BRANCH
107	[18,167]	[38,185]	34.899	BRANCH	BRANCH
108	[1,170]	[18,167]	18.071	TIP	BRANCH
109	[87,167]	[88,170]	2.414	TIP	BRANCH
110	[88,170]	[108,172]	19.828	BRANCH	BRANCH
111	[88,170]	[71,175]	17.657	BRANCH	BRANCH
112	[109,167]	[108,172]	4.414	TIP	BRANCH
113	[120,175]	[108,172]	11.828	TIP	BRANCH
114	[57,174]	[66,176]	8.828	BRANCH	BRANCH
115	[57,174]	[66,176]	17.657	BRANCH	BRANCH
116	[71,175]	[66,176]	4.414	BRANCH	BRANCH
117	[71,175]	[116,188]	49.971	BRANCH	BRANCH
118	[27,188]	[38,185]	11.243	TIP	BRANCH
119	[38,185]	[47,197]	15.485	BRANCH	BRANCH
120	[208,187]	[213,186]	4.414	TIP	BRANCH
121	[216,183]	[213,186]	3.414	TIP	BRANCH
122	[238,181]	[213,186]	31.556	TIP	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
123	[162,215]	[116,188]	57.355	TIP	BRANCH
124	[166,171]	[116,188]	56.799	TIP	BRANCH
125	[485,189]	[481,193]	5.243	TIP	BRANCH
126	[477,191]	[481,193]	3.828	TIP	BRANCH
127	[481,195]	[481,193]	1.0	TIP	BRANCH
128	[68,195]	[47,197]	20.828	TIP	BRANCH
129	[47,197]	[53,209]	13.485	BRANCH	BRANCH
130	[286,202]	[284,203]	1.414	TIP	BRANCH
131	[283,209]	[284,203]	5.414	TIP	BRANCH
132	[284,195]	[284,203]	7.414	TIP	BRANCH
133	[53,209]	[86,216]	37.142	BRANCH	BRANCH
134	[51,212]	[53,209]	2.414	TIP	BRANCH
135	[86,227]	[86,216]	11.657	TIP	BRANCH
136	[103,215]	[86,216]	16.828	TIP	BRANCH
137	[47,219]	[43,229]	10.657	TIP	BRANCH
138	[43,229]	[30,242]	19.728	BRANCH	BRANCH
139	[55,235]	[43,229]	13.071	TIP	BRANCH
140	[15,237]	[7,237]	7.828	TIP	BRANCH
141	[7,237]	[7,244]	6.0	BRANCH	BRANCH
142	[12,218]	[7,237]	23.899	TIP	BRANCH
143	[21,241]	[30,242]	8.414	TIP	BRANCH
144	[30,242]	[7,244]	29.899	BRANCH	BRANCH
145	[0,254]	[7,244]	12.485	TIP	BRANCH
146	[95,248]	[95,255]	6.0	TIP	BRANCH
147	[93,257]	[95,255]	1.414	TIP	BRANCH
148	[99,257]	[95,255]	3.414	TIP	BRANCH
149	[79,266]	[53,261]	27.071	TIP	BRANCH
150	[48,323]	[53,261]	73.113	TIP	BRANCH
151	[43,240]	[53,261]	23.728	TIP	BRANCH
152	[157,280]	[154,279]	2.414	TIP	BRANCH
153	[154,283]	[154,279]	3.0	TIP	BRANCH
154	[150,274]	[154,279]	5.243	TIP	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
155	[471,418]	[484,417]	14.657	TIP	BRANCH
156	[511,398]	[484,417]	35.627	TIP	BRANCH
157	[484,417]	[449,440]	44.698	BRANCH	BRANCH
158	[75,420]	[59,424]	16.657	TIP	BRANCH
159	[59,424]	[30,427]	30.071	BRANCH	BRANCH
160	[62,433]	[59,424]	9.243	TIP	BRANCH
161	[33,419]	[30,427]	8.243	TIP	BRANCH
162	[16,433]	[30,427]	15.071	TIP	BRANCH
163	[102,425]	[106,430]	6.243	TIP	BRANCH
164	[111,430]	[106,430]	4.0	TIP	BRANCH
165	[106,436]	[106,430]	5.0	TIP	BRANCH
166	[449,440]	[444,441]	4.414	BRANCH	BRANCH
167	[485,443]	[449,440]	39.142	TIP	BRANCH
168	[442,438]	[444,441]	2.828	TIP	BRANCH
169	[444,441]	[421,478]	45.527	BRANCH	BRANCH
170	[163,444]	[151,451]	13.899	TIP	BRANCH
171	[130,458]	[151,451]	22.899	TIP	BRANCH
172	[150,454]	[151,451]	2.414	TIP	BRANCH
173	[332,461]	[332,463]	1.0	TIP	BRANCH
174	[307,465]	[332,463]	25.657	TIP	BRANCH
175	[82,446]	[75,477]	33.728	TIP	BRANCH
176	[82,479]	[75,477]	6.828	TIP	BRANCH
177	[72,480]	[75,477]	3.828	TIP	BRANCH
178	[348,452]	[421,478]	85.255	TIP	BRANCH
179	[427,509]	[421,478]	32.485	TIP	BRANCH
180	[57,452]	[56,489]	40.556	TIP	BRANCH
181	[58,489]	[56,489]	1.0	TIP	BRANCH
182	[42,490]	[56,489]	14.243	TIP	BRANCH
183	[110,0]	[134,10]	28.728	TIP	TIP
184	[260,0]	[273,10]	18.899	TIP	TIP
185	[280,18]	[277,56]	43.385	TIP	TIP
186	[511,22]	[511,57]	35.828	TIP	TIP

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
187	[35,35]	[33,51]	19.899	TIP	TIP
188	[64,40]	[50,46]	16.485	TIP	TIP
189	[441,58]	[414,63]	29.071	TIP	TIP
190	[64,59]	[46,76]	30.899	TIP	TIP
191	[175,66]	[172,82]	18.071	TIP	TIP
192	[267,68]	[281,87]	24.799	TIP	TIP
193	[430,79]	[394,109]	53.698	TIP	TIP
194	[468,113]	[426,122]	45.728	TIP	TIP
195	[108,129]	[115,145]	33.385	TIP	TIP
196	[399,141]	[389,157]	20.142	TIP	TIP
197	[51,143]	[57,150]	10.071	TIP	TIP
198	[170,143]	[194,174]	40.941	TIP	TIP
199	[232,172]	[222,176]	11.657	TIP	TIP
200	[185,187]	[177,197]	17.314	TIP	TIP
201	[342,197]	[341,224]	28.243	TIP	TIP
202	[400,205]	[394,213]	11.657	TIP	TIP
203	[110,212]	[131,234]	33.87	TIP	TIP
204	[167,222]	[190,236]	28.799	TIP	TIP
205	[88,231]	[97,241]	13.728	TIP	TIP
206	[226,231]	[236,231]	10.0	TIP	TIP
207	[491,241]	[509,242]	18.414	TIP	TIP
208	[352,269]	[398,271]	50.142	TIP	TIP
209	[9,270]	[32,293]	36.042	TIP	TIP
210	[215,290]	[195,310]	31.213	TIP	TIP
211	[161,298]	[178,308]	21.728	TIP	TIP
212	[273,298]	[274,310]	13.243	TIP	TIP
213	[0,310]	[22,317]	25.728	TIP	TIP
214	[450,313]	[386,327]	71.456	TIP	TIP
215	[166,320]	[176,325]	12.071	TIP	TIP
216	[93,334]	[78,337]	17.899	TIP	TIP
217	[49,339]	[11,343]	43.657	TIP	TIP
218	[253,344]	[217,348]	54.385	TIP	TIP

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
219	[15,357]	[38,367]	27.142	TIP	TIP
220	[181,363]	[183,371]	9.414	TIP	TIP
221	[291,374]	[277,378]	15.657	TIP	TIP
222	[97,378]	[104,398]	23.728	TIP	TIP
223	[415,384]	[414,407]	24.243	TIP	TIP
224	[88,399]	[99,422]	32.87	TIP	TIP
225	[226,488]	[235,510]	27.728	TIP	TIP
226	[19,507]	[32,511]	14.657	TIP	TIP